Spring-18

Group-B

(4) (A) By the sum rule it follows that there are ~~11!~~ $\frac{9!}{3!(9-3)!} + \frac{11!}{4!(11-4)!} = 414$ Ans

4 (B)

A Principle of inclusion - exclusion can be generalized to find the number of ways to do one of n different task or, equivalently, to find the number of elements in ~~#~~ the union of n sets, ~~wheever~~ whenever n is a positive integer.

4(B) 2nd Part

Since each of the 8·10 bits is is either a 0 or a 1 the answer is,

$2^7 + 2^8 - 2^5 = 352$

Ans

0	0	0	.	.	.	.	.	.	.

$2^7 = 128$

.	.	.	.	.	.	.	.	0	0

$2^8 = 256$

0	0	0	.	.	.	.	.	0	0

$2^5 = 32$

4(c)

Since each of Choose ~~made~~ can be 3 different type of ice-creams.

3	3	3	3	3	3	3

the ans is = 3^7 = ~~20~~ 2187

Ans

Autumn-18

Group-B

4(b)

Since SUCCESS have seven letters where are threes Ss, two ~~Es~~ Cs, one U, and one E.

By the Product rule, the number of different String.

$C(7,3)\ C(4,2)\ C(2,1)\ C(1,1)$

$$= \frac{7!}{3!\,4!} \cdot \frac{4!}{2!\,2!} \cdot \frac{2!}{1!\,1!} \cdot \frac{1!}{1!\,0!}$$

$$= \frac{7!}{3!\,2!\,1!\,1!} = 420$$

4(c) Let P be the total number of password and ~~P~~ let P_5, P_6, be the password of lenght 5, 6

~~P~~

~~P~~ By the sum role $P = P_5 + P_6$

CamScanner

$P_5 = 36^5 - 26^5 = 48584800$

$P_6 = 36^6 - 26^6 = 1,867,866,560$

Consequently, $P = P_5 + P_6$

$= \cancel{3,735,733,120}$

$= 19,16451360$

d]

Spring-19

Principles of Counting

- *Combinatorics*, the study of arrangements of objects, is an important part of discrete mathematics. This subject was studied as long ago as the seventeenth century.
- Enumeration, counting of objects with certain properties, is an important part of combinatorics.
- Counting is used to solve many different types of problems. For example, counting is used to determine the complexity of algorithms. Counting is also required to determine whether there are enough telephone numbers or Internet protocol addresses to meet demand. Furthermore, counting techniques are used extensively when probabilities of events are computed.
- We will study here: Basic rules of counting, Pigeonhole principle, Permutations and Combinations, Analyzing gambling games, Computer Simulations etc.

Basic Counting Principles: The Sum Rule

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2, where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.

The Generalized principle of Inclusion-Exclusion:

- The principle of inclusion-exclusion can be generalized to find the number of ways to do one of n different tasks or, equivalently, to find the number of elements in the union of n sets, whenever n is a positive integer.
- Let $A_1, A_2, \ldots, A_n$ be finite sets. Then
- $| A_1 \cup A_2 \cup \ldots \cup A_n | = \sum | A_i | - \sum | A_i \cap A_j | + \sum | A_i \cap A_j \cap A_k | - \ldots \ldots + (-1)^{n+1} | A_1 \cap A_2 \cap \ldots \ldots \cap A_n |$
-
- *Example*: Give a formula for the number of elements in the union of four sets.
- *Solution*: The inclusion-exclusion principle shows that
- $| A_1 \cup A_2 \cup \ldots \cup A_n | = |A_1| + |A_2| + |A_3| + |A_4| - |A_1 \cap A2| - |A_1 \cap A_3| - |A_1 \cap A_4| - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_3 \cap A_4| + | A_1 \cap A_2 \cap A_3 | + | A_1 \cap A_2 \cap A4 | + | A_1 \cap A_3 \cap A_4 | + | A_2 \cap A_3 \cap A_4 | - | A_1 \cap A_2 \cap A_3 \cap A_4|.$
- NB: Note that this formula contains 15 different terms i.e. $2^4 - 1$.